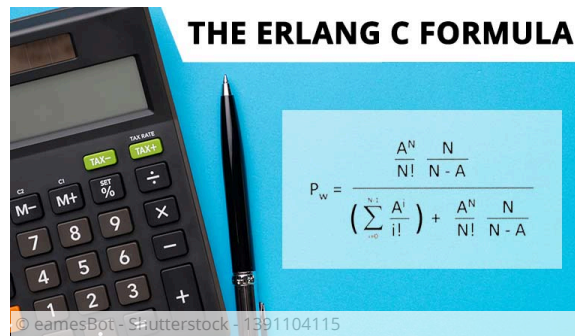


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Erlang C Formula – Made Simple With an Easy Worked Example



[in](#) [X](#) [✉](#) [👁](#) 472,170

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[Occupancy](#), [Service Level](#), [Shrinkage](#),...



The Erlang C formula is quite complex to look at, but can be straightforward to apply, if you follow these easy steps.

We have also included a [worked example in Excel](#) to help you follow the maths.

But before we dive into the maths, here is some key information about

the Erlang C formula and its derivation:

What Is the Erlang C Formula?

The Erlang C formula is a mathematical equation for calculating the number of agents (advisors) that you need in a call centre, given the number of calls and the service level that you want to achieve.

The Erlang C formula is the most important part of the equation. It allows you to work out the probability that a call waits (P_w), given the Traffic Intensity (A) and the Number of Agents (N) available.

$$P_w = \frac{\frac{A^N}{N!} \frac{N}{N-A}}{\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right) + \frac{A^N}{N!} \frac{N}{N-A}}$$

It was named after the Danish Mathematician A.K. Erlang, who developed the Erlang C formula 100 years ago in 1917.

A.K. Erlang published (in Danish) his landmark paper "Solution of some problems in the theory of probabilities of significance in automatic telephone exchanges," In the Journal Electroteknikeren 13: 5-13 (1917)

It first appeared in English in the
Post Office Electrical Engineers
Journal Volume 10 (1917–1918)
Pages 189–197

It is a derivation of the Poisson
Formula – but using a special case
where $k=0$. This assumes that the
Average Handling Time of calls is
distributed by a version of Poisson
called the [Erlang Distribution](#).

For more information on this
topic, read our article: [How Is
Average Handling Time
Distributed? It Is Not How You
Think!](#)

Why Should You Use Erlang C?

As a call centre manager, if you
want to save costs and improve
your customer experience (CX), you
should understand Erlang C and
how to use it.

Why? By understanding the
principles behind Erlang C, call
centre managers can better:

- Understand call load and call distribution
- Understand service level
- Manage costs
- Create a more stable user experience

After all, you don't want too many agents to be scheduled when there are only a few calls.



A Step by Step Guide to the Erlang C Formula

1. The Inputs

Before we start, we need a number of inputs.

We will use a worked example of 100 calls per half hour, with an Average Handling Time of 3 minutes. I have chosen this example as it uses nice round numbers in the maths.

We will use industry averages for Service Level, Shrinkage and Maximum Occupancy, which are shown below:

- Number of calls – 100
- In a period of minutes – 30
- Average Handling Time (seconds) – 180 (3 minutes)
- Required Service Level – 80%
- Target Answer Time (Seconds) – 20

- Maximum Occupancy – 85%
- Shrinkage – 30%

2. Work Out the Number of Calls Per Hour

So, we have 100 calls per 30 minutes, which works out as 200 calls per hour

3. Work Out the Traffic Intensity (A)

The traffic intensity is the length of time that all of the phone calls would take if ordered end to end.

So if we have 200 calls with an Average Handling Time of 3 minutes, we would have a total of $200 \times 3 = 600$ call minutes.

To work out the traffic intensity, take the call minutes and divide by 60 to get the number of call hours.

So, 600 call minutes / 60 = 10 Call Hours.

Now the technical unit for Call Hours is called an Erlang.

So the traffic intensity = 10 Call Hours = 10 Erlangs.

4. Estimate the Raw Number of Agents Required (N)

We now need to estimate the Raw Number of Agents Required to handle this traffic intensity.

We know that we have 10 Erlangs (10 calls hours of traffic per hour). This means that the minimum number of agents that we would need in the call centre would be 10 agents.

This figure of 10 agents would assume that all calls arrive at exactly the moment that a previous call has finished and that there is no queuing.

So, we start with adding 1 to the traffic intensity.

**Estimate 1: $N = A + 1 = 10 + 1 = 11$
Agents**

We then feed the Traffic Intensity (A) and the Number of Agents (N) into the Erlang C formula to see what is the probability that a call waits and then work out the Service Level.

We then increase the Number of Agents until Service Level is met.

5. Calculate the Erlang Formula for Probability a Call Waits

Here is the formula from Wikipedia:

$$P_w = \frac{\frac{A^N}{N!} \frac{N}{N-A}}{\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right) + \frac{A^N}{N!} \frac{N}{N-A}}$$

You may also see it written like this.

$$P_w = \frac{\frac{u^m}{m!}}{\frac{u^m}{m!} + (1 - \rho) \left(\sum_{k=0}^{N-1} \frac{u^k}{k!} \right)} \quad \text{where } \rho = \frac{u}{m}$$

Don't worry; this is the same formula. It just uses different units (u) and (m) and has a factor multiplied across the top and the bottom of the formula. (If you are uncertain, scroll down to the bottom of the page where I will show how they are the same formula.)

This looks scary so let's look at the key elements individually and build it up in stages.

6. Work Out N! (N Factorial)

N! is mathematical notation for N factorial.

So you need to cast your mind back to school maths. A factorial is the product of an integer (whole number) and all the integers (whole numbers) below it.

For example $3! = 3 \text{ Factorial} = 3 \times 2 \times 1 = 6$

So $N! = 11! = 11 \text{ Factorial} = 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 39916800$

For reference, here is a lookup table of the first 20 Factorials. We will need this for the worked example.

Number	Factorial
N	N!
1	1
2	2
3	6
4	24
5	120
6	720
7	5040
8	40320
9	362880
10	3628800
11	39916800
12	479001600
13	6227020800
14	87178291200
15	1.30767×10^{12}
16	2.09228×10^{13}
17	3.55687×10^{14}
18	6.40237×10^{15}
19	1.21645×10^{17}
20	2.43290×10^{18}

Luckily, Excel has a formula that can do this for you.

So, for 11 Factorial it is **=FACT(11)**

N! = 11! = 11 Factorial = FACT (11) = 39916800

7. Be Careful With Large Factorials

Note that factorials get very large as you increase the number of agents and very soon can cause Excel (and some computer languages) to overload, as highlighted below.

10! = 10 Factorial = FACT(10) = 3628800

100! = 100 Factorial = FACT(100) = 9.3326×10^{157}

170! = 170 Factorial = FACT(170) = 7.2574×10^{306}

171! = 171 Factorial = FACT(171) = #NUM!

The **#NUM!** shows that Excel is overloaded and can give you erroneous answers.

On some Erlang Calculators you will see this Error written as **NaN** (Not a Number).

So it won't easily work in Excel above 170 agents.

Luckily, we have devised a Fast Erlang formula for a high number

of agents as part of our [Erlang Calculator](#)

8. Work Out the Powers – A^N

So, A^N means A multiplied by A, N times.

For example: 2 to the power of 2 =
 $2^2 = 2 \times 2 = 4$

So if $A = 10$ and $N = 11$, then $A^N = 10^{11} = 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$

Luckily, Excel has a formula for this
= POWER (10,11)

$A^N = 10^{11} = \text{POWER}(10,11) = 1 \times 10^{11}$

9. Let's Simplify the Erlang C Formula

We can use a substitution, to substitute X and Y for the complex parts of the equation: $P_w = X / (Y + X)$

$$\text{If } X = \frac{A^N}{N!} \frac{N}{N-A} \quad \text{And } Y = \left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right)$$

$$P_w = \frac{\frac{A^N}{N!} \frac{N}{N-A}}{\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right) + \frac{A^N}{N!} \frac{N}{N-A}} = \frac{X}{Y + X}$$

10. Let's Work Out the Top Row of the Erlang Formula (X)

$$\frac{A^N}{N!} \frac{N}{N-A}$$

$$X = A^N / N! * [N / (N-A)]$$

$$X = 10^{11} / 11! * [11 / (11-10)]$$

$$X = 10^{11} / 11! * (11 / 1)$$

$$X = 10^{11} / 11! * 11$$

$$X = 27557.32$$

11. Work Out the Sum of a Series (Y)

$$\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right)$$

Now this looks scary, but it is simply adding together a sum of $A^i / i!$ from zero to a maximum value of $N-1$.

In this case, you start with $i=0$ and keep going through until $i= N-1$

So, if $N = 11$, $N - 1 = 10$ and we loop through from $i= 0$ until $i = 10$, as shown below.

$$\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right) = \frac{A^0}{0!} + \frac{A^1}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \frac{A^5}{5!} + \frac{A^6}{6!} + \frac{A^7}{7!} + \frac{A^8}{8!} + \frac{A^9}{9!} + \frac{A^{10}}{10!}$$

Σ is the mathematical symbol for the sum of a series, so you add together all of the results of the individual loops.

i	i!	A ⁱ	A ⁱ /i!	Σ A ⁱ /i!
---	----	----------------	--------------------	----------------------

0	1	1	1	1
1	1	10	10	11
2	2	100	50	61
3	6	1000	166. 7	227.7
4	24	10000	416. 7	644.3
5	120	100000	833. 3	1477. 7
6	720	1000000	1388 .9	2866. 6
7	5040	10000000	1984 .1	4850. 7
8	40320	10000000 0	2480 .2	7330. 8
9	36288 0	10000000 00	2755 .7	10086 .6
10	36288 00	10000000 000	2755 .7	12842 .3

This means that the sum of the series **Y = 12842.3**

12. Put X and Y into the Erlang C Formula (The probability a call has to wait)

So $P_w = X / (Y + X)$

$P_w = 27557 / (12842 + 27557) =$
0.6821

Or in percentage terms 68.2%

13. Calculate the Service Level

Service Level follows the formula:

$$\text{Service Level} = 1 - \left[P_w \times e^{-[(N - A) \times (\text{Target Time} / \text{AHT})]} \right]$$

Where e is the mathematical constant (Euler's number) and will always equal 2.71828.

Luckily, Excel has an equation for this =EXP()

Let's work out $-(N - A) \times (\text{TargetTime} / \text{AHT})$

Let's work out $-(11 - 10) \times (20 / 180)$
= -0.111

Then work out Service Level

$SL = 1 - (0.6821 \times \text{EXP}(-0.11111)) =$
0.390

Service Level as a Percentage =
39.0%

This is below the target of 80%, so we need to increase the number of agents.

14. Increase the Number of Agents (N) by 1 to See if Service Level Is Achieved

We keep on increasing the Number of Agents (N) by 1 to see if Service Level is achieved. We keep looping through steps 4 to 12 until we meet service level.

This can be done by hand but can be a bit tedious.

For the sake of completeness, I have included the calculations below:

14.1 Number of Agents $N = 12$

Traffic Intensity $A = 10$

Work out $X = 12526.1$

i	$i!$	A^i	$A^i/i!$	$\Sigma A^i/i!$
0	1	1	1	1
1	1	10	10	11
2	2	100	50	61
3	6	1000	166. 7	227.7
4	24	10000	416. 7	644.3
5	120	100000	833. 3	1477. 7
6	720	1000000	1388 .9	2866. 6
7	5040	10000000	1984 .1	4850. 7
8	40320	10000000 0	2480 .2	7330. 8
9	36288 0	10000000 00	2755 .7	1008 6.6
1 0	36288 00	10000000 000	2755 .7	1284 2.3
1 1	39916 800	10000000 0000	2505 .2	1534 7.5

Work out $Y = 15347.5$

Probability Call Waits = 0.4494

Service Level = 0.640

Service Level as a Percentage

64.0%. This is below the target of 80% so we need to increase the number of agents.

14.2. Increase the Number of Agents by 1 to See if Service Level Is Achieved

Number of Agents $N = 13$

Traffic Intensity $A = 10$

Work out $X = 6958.9$

i	$i!$	A^i	$A^i/i!$	$\Sigma A^i/i!$
0	1	1	1	1
1	1	10	10	11
2	2	100	50	61
3	6	1000	166. 7	227.7
4	24	10000	416. 7	644.3
5	120	100000	833. 3	1477. 7
6	720	1000000	138 8.9	2866. 6
7	5040	10000000	198 4.1	4850. 7
8	40320	10000000 0	248 0.2	7330. 8
9	362880	10000000 00	275 5.7	1008 6.6
10	362880 0	10000000 000	275 5.7	1284 2.3

1	399168	10000000	250	1534
1	00	0000	5.2	7.5
1	479001	10000000	208	1743
2	600	00000	7.7	5.2

So value of $Y = 17435.2$

Probability Call Waits 0.2853

Service Level 0.796

Service Level as a Percentage 79.6%

This is below the target of 80% so we need to increase the number of agents

14.3. Increase the Number of Agents by 1 to See if Service Level Is Achieved

Number of Agents = $N = 14$

Traffic Intensity = $A = 10$

Work out $X = 4014.8$

i	$i!$	A^i	$A^i/i!$	$\Sigma A^i/i!$
0	1	1	1	1
1	1	10	10	11
2	2	100	50	61
3	6	1000	166. 7	227. 7
4	24	10000	416. 7	644. 3
5	120	100000	833. 3	1477 .7

6	720	1000000	138 8.9	2866 .6
7	5040	10000000	198 4.1	4850 .7
8	40320	100000000	248 0.2	7330 .8
9	362880	100000000 0	275 5.7	1008 6.6
1 0	362880 0	100000000 00	275 5.7	1284 2.3
1 1	399168 00	100000000 000	250 5.2	1534 7.5
1 2	479001 600	100000000 0000	208 7.7	1743 5.2
1 3	622702 0800	100000000 00000	160 5.9	1904 1.1

So the value of Y = **19041.1**

Probability Call Waits = 0.1741

Service Level = 0.888

Service Level as a Percentage =
88.8%

This is **above** the target of 80%

So the Raw Number of Agents = 14

*(Please note that this does not
include Shrinkage or Maximum
Occupancy)*

15. Average Speed of Answer (ASA)

$$\text{Average Speed of Answer (ASA)} = \frac{P_w \times \text{Average Handling Time}}{(\text{No. of Agents} - \text{Traffic Intensity})}$$

$$\text{ASA} = 0.1741 \times 180 / (14 - 10) = 7.8$$

seconds

16. Percentage of Calls Answered Immediately

$$\text{Immediate Answer} = (1 - P_w) \times 100$$

$$\text{Immediate Answer} = (1 - 0.1741) = 0.826 = 82.6\%$$

17. Check Maximum Occupancy

Occupancy is the percentage of time that all call centre agents spend handling customer contacts. The formula is shown below.

$$\text{Occupancy} = \frac{\text{Traffic Intensity (Erlangs)}}{\text{Raw Agents}} \times 100$$

$$\text{Raw Agents} = 14$$

$$\text{Traffic Intensity } A = 10$$

$$\text{Occupancy} = (10 / 14) \times 100 = 71.4\%$$

This is less than 85% so we can keep it at this.

If this is more than 85%, the number of Raw Agents = Traffic Intensity / (Occupancy %/100)

If you have an occupancy of more than 85%, your agents will likely get burnt out and you will find that Average Handling Time (AHT) may rise to cover it.

18. Factor In Shrinkage

Shrinkage is a factor that is widely used in the industry to include holidays, sickness, training and meetings etc. The industry average is around 30–35%

Shrinkage = 30%

$$\text{No. of Agents Required} = \frac{\text{Raw Agents}}{[1 - (\text{Shrinkage} / 100)]}$$

Number of Agents Required = $14 / 0.7 = 20$

To find out your contact centre's shrinkage rate, read our article: [How to Calculate Contact Centre Shrinkage](#)

19. Summary

So the total number of Agents Required – **20**

Service Level – **88.8%**

Probability a call has to wait – **17.4%**

Average Speed of Answer – **7.8 Seconds**

% of calls Answered Immediately – **82.6%**

20. There Is an Easier (and Faster) Way to Do an Erlang C Calculation

There is a much easier (and faster) way to use the Erlang C Formula.

We have developed two Erlang Calculators

- [Erlang Calculator \(Online Version\)](#)
- [Excel Erlang Calculator \(Handy spreadsheet format\)](#)

The added advantage of these calculators is that we have already figured out how to increase the calculations above the limit of 172 Agents and they are accurate to 10,000 Agents for the online Version and 600 Agents for the Excel Version.

Also, if you want a [spreadsheet that explains the Erlang C Formula you can download it here](#).



Proof That the Two Versions of the Erlang C Formula Are the Same

Here is the Wikipedia Version.

$$P_w = \frac{\frac{A^N}{N!} \frac{N}{N-A}}{\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right) + \frac{A^N}{N!} \frac{N}{N-A}}$$

Here is the other version.

$$P_w = \frac{\frac{u^m}{m!}}{\frac{u^m}{m!} + (1 - \rho) \left(\sum_{k=0}^{N-1} \frac{u^k}{k!} \right)} \quad \text{where } \rho = \frac{u}{m}$$

Let's substitute $\rho = u/m$.

$$P_w = \frac{\frac{u^m}{m!}}{\frac{u^m}{m!} + \left(1 - \frac{u}{m}\right) \left(\sum_{k=0}^{N-1} \frac{u^k}{k!} \right)}$$

Now since u is traffic intensity (A), and m is the number of agents (n) we can substitute the equation.

$$P_w = \frac{\frac{A^N}{N!}}{\frac{A^N}{N!} + \left(1 - \frac{A}{N}\right) \left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right)}$$

We can write the numerators and denominators of part of the equation in a different way.

$$\text{Now } \left(1 - \frac{A}{N}\right) = \frac{N}{N} - \frac{A}{N} = \frac{N - A}{N}$$

Let's divide the top half and the bottom half of the equation by $(N - A) / N$

$$P_w = \frac{\frac{A^N}{N!}}{\frac{A^N}{N!} + \frac{N - A}{N} \left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right)}$$

Divide top and bottom by $\frac{N - A}{N}$

Which produces

$$P_w = \frac{\frac{A^N}{N!} \frac{N-A}{N}}{\frac{A^N}{N!} \frac{N-A}{N} + \left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right)}$$

Now since dividing by a/b is the same as multiplying by b/a , dividing by $(N-A)/N$ is the same as multiplying by $N/(N-A)$.

Which gives

$$P_w = \frac{\frac{A^N}{N!} \frac{N}{N-A}}{\frac{A^N}{N!} \frac{N}{N-A} + \left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right)}$$

And since $X + Y$ is the same as $Y + X$ we can express it as

$$P_w = \frac{\frac{A^N}{N!} \frac{N}{N-A}}{\left(\sum_{i=0}^{N-1} \frac{A^i}{i!} \right) + \frac{A^N}{N!} \frac{N}{N-A}}$$

Which is our original Erlang C formula.

Find out more about the other Erlang equations and functions in our articles:

- [A Beginner's Guide to the Erlang A Formula](#)
- [What Is an Erlang Calculator?](#)
- [What are Erlangs?](#)

Author: [Jonty Pearce](#)

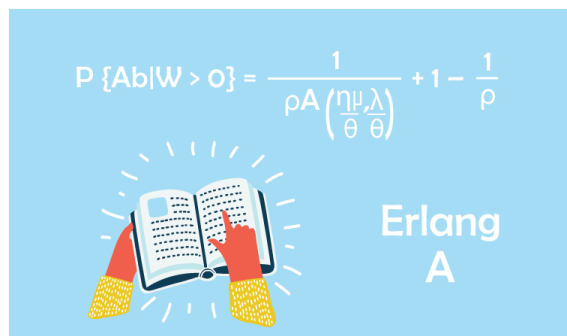
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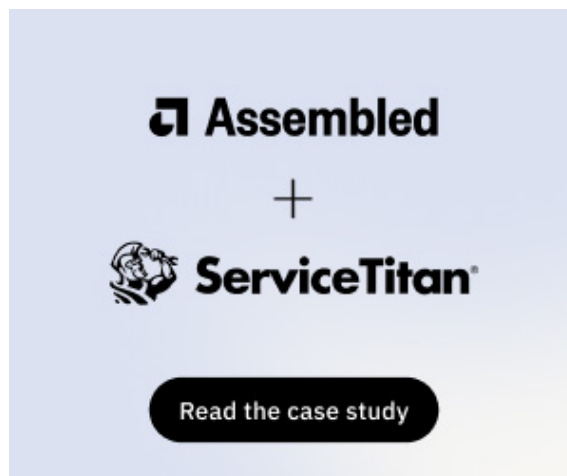
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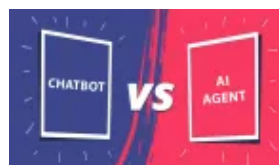
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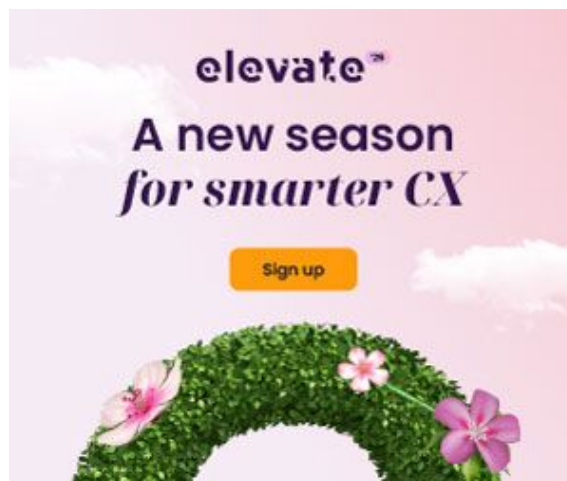
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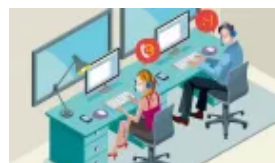
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